Yuxuan Ma

M.S. Student, The University of Hong Kong

Available to start a full-time Research Assistant position immediately

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Research Interests

Human-centered evaluation of AI systems: how people interpret, verify, and appropriately rely on the outputs of computer vision models and data-driven decision-support tools. Grounded in my experience with Vision Transformer architectures, object detection, and business data analytics, I aim to study interface mechanisms—explanations, uncertainty displays, and interactive visualizations—that make AI-assisted decision-making more transparent and trustworthy.

Education

Sep 2025 – Nov 2026 (expected)
Hong Kong, China

The University of Hong Kong
M.S. in Engineering, Innovative Design and Technology, with Distinction
Thesis: Feature Aggregation in Vision Transformers for CIFAR-10 Image Classification

Sep 2018 – Jun 2023
Beijing, China

Communication University of China
B.Eng. in Computer Science and Technology
Thesis: YOLOv5-based Intelligent Traffic System

Research and Project Experience

Apr 2026 – Aug 2026
Hong Kong, China

The Hong Kong Polytechnic University
Research Assistant, Prof. Rongpan Hu's Group
Reliability of AI Defect Detection for Infrastructure Inspection

- Diagnosed why AI crack detection becomes unreliable on model-generated detections: across 554 matched instances, ruled out sample selection as a confound and isolated proposal framing as the dominant degradation pathway (false-alarm rate at 95% recall rose from 5% to 43%).
- Built a nine-level controlled expansion benchmark with image-clustered bootstrap confidence intervals, identifying an early-collapse regime where reliability degraded sharply at low target occupancy before plateauing, i.e. where AI outputs most warrant human review.
- Translated near-OOD false alarms into selective-review workload, benchmarking reliability-ranked triage against random and Oracle baselines over 106 candidates, motivated by reliability requirements in real-world bridge inspection.

Mar 2026 – Jul 2026
Shenzhen, China

Beihang University Shenzhen Institute
Research Assistant, Prof. Baojiang Wang's Group
Uncertainty-Aware Human-AI Collaborative Inspection for Power-Grid Insulators

- Designed an uncertainty-aware interface for UAV-based power-grid insulator inspection, using confidence- and risk-based visual cues to prioritize uncertain cases for manual review and support calibrated reliance on AI.
- Designed and implemented an explainable-AI module linking predictions to supporting image regions, and iteratively refined information hierarchy and interaction flow to streamline interpretation of inspection results.

2026
Hong Kong, China

The University of Hong Kong
Master's Thesis Research, Feature Aggregation in Vision Transformers
Supervisor: Prof. Grantham Pang

- Compared three Vision Transformer feature-aggregation mechanisms—learnable CLS token, fixed CLS token, and mean pooling—on CIFAR-10 under controlled training settings.
- Found mean pooling achieved the highest test accuracy of 80.16%, improving the CLS-token baseline by 1.05 percentage points.

Nov 2022 – Mar 2023
Beijing, China

Communication University of China
Undergraduate Thesis Research, Vision-Based Intelligent Traffic System

- Selected YOLOv5s after benchmarking Faster R-CNN, SSD, and YOLOv5 for VisDrone-based traffic-scene detection.
- Built a PyQt5, Python, and OpenCV system for image, video, and live-camera vehicle detection, tracking, flow counting, and speed estimation.

May – Jun 2026 **Tmall Beauty Strategy Dashboard**

- Built a multi-page Power BI dashboard on Tmall Double 11 beauty-category data, with a Power Query and DAX metrics layer covering revenue, volume, average selling price, price bands, and consumer segments.
- Combined competitor benchmarking, product-keyword analysis, and pre/post-campaign discount analysis to identify skincare as the primary revenue driver and inform pricing and assortment strategy.

Aug – Sep 2025 **China Tourist Attractions Data Analysis and Visualization**

- Collected, cleaned, and analyzed tourism data across 34 Chinese provincial-level regions, using correlation analysis and Tableau dashboards to identify demand drivers and visualize regional tourism patterns.

Professional Experience

Oct 2025 – Feb 2026 **Deloitte**
Data Analytics Intern

- Developed AR/ARIMA time-series models with seasonal decomposition to forecast quarterly revenue growth and quantify forecast uncertainty via prediction intervals.
- Built a five-region feature framework covering share, growth, and contribution, and used key-factor selection to cut forecast error by about 38% over the baseline.
- Consolidated forecasting, validation, and regional-driver analysis into an analytical report for the project team.

Jul 2023 – Jul 2025
Jinan, China

Jinan Ousi Software Co., Ltd.
Software Development Engineer

- Delivered production WMS and ERP systems (Spring Boot, MyBatis, RabbitMQ, Spring Cloud); containerized services with Docker and migrated part of the business to Qiniu Cloud.

Jun 2022 – Nov 2022
Beijing, China

Beijing Bole Technology Co., Ltd.
Server-side Development Engineering Intern

- Optimized back-end functions for two live game titles, deployed and maintained servers, and used Redis caching and Qiniu Cloud object storage to reduce load times.

Dec 2021 – May 2022
Beijing, China

Beijing iHandy Mobile Inc.
Assistant Software Engineer

- Contributed to advertising and back-end cloud-service platforms, co-designing database structures and building JDBC-based data-access layers for core business logic.

Technical and Methodological Skills

Programming and Tools: Python, SQL, Java, C++, JavaScript, Power BI, DAX, Power Query, Tableau, Excel, MySQL, Hive
AI and Methods: Vision Transformers, YOLOv5, OpenCV, PyQt5, pandas, NumPy, scikit-learn, EDA, ARIMA/SARIMA forecasting, A/B testing, ETL, model evaluation, data visualization
Languages: Mandarin Chinese (native); English (fluent)
Certifications: Big Data Analyst (Advanced), Education & Examination Center, Ministry of Industry and Information Technology, China (Aug 2025)